StellarShift-Bench

Reliability under real cross-survey shift in stellar spectroscopy

1,088	+108-242%	51
model-held-out DESI evaluation stars	source-only MAE degradation	passing tests

Executive finding. A model that is accurate on held-out LAMOST spectra degrades sharply on disjoint DESI DR1 spectra even when both domains use the exact APOGEE DR12 ASPCAP v603 reference-label scale. The transferred model also becomes overconfident. Labeled target retraining recovers much of the loss; unlabeled CORAL does not and materially harms metallicity.

What changed in v1.2.3

- Replaced the stale standalone notebook with the release-built version and made every code cell runnable against archived evidence.
- Added an independent two-sample star bootstrap for source-to-target MAE change and auditable source-holdout predictions on full reruns.
- Added the missing APOGEE-to-DESI crossmatch builder and the complete 4,662,192 to 1,827 to 1,576 to 1,555 selection audit.
- Reframed the release as prespecified quality-gated and retained the deterministic v1.2.2 release and Colab safeguards.

Rama Serhan | Technical report | 16 August 2026

1. Study design and prespecified quality gate

The source consists of public high-S/N LAMOST DR2 optical spectra from the Ho et al. tutorial. The target consists of APOGEE DR12 stars position-matched within 1.5 arcsec to primary DESI DR1 spectra and retrieved through SPARCL. APOGEE_ID is the cross-survey identity key; source-target overlap is removed before splitting.

Partition	Observed	Gate	Purpose
LAMOST source train	1,032	>=1,000	Feature/model fitting
LAMOST source holdout	259	>=200	In-domain error and calibration
DESI target adaptation	467	>=100	Adaptation/recalibration only
DESI target evaluation	1,088	>=350	model-held-out final evaluation
Evaluation giants	1,024	>=50	Subgroup support
Evaluation metal-poor	55	>=50	Out-of-support visibility

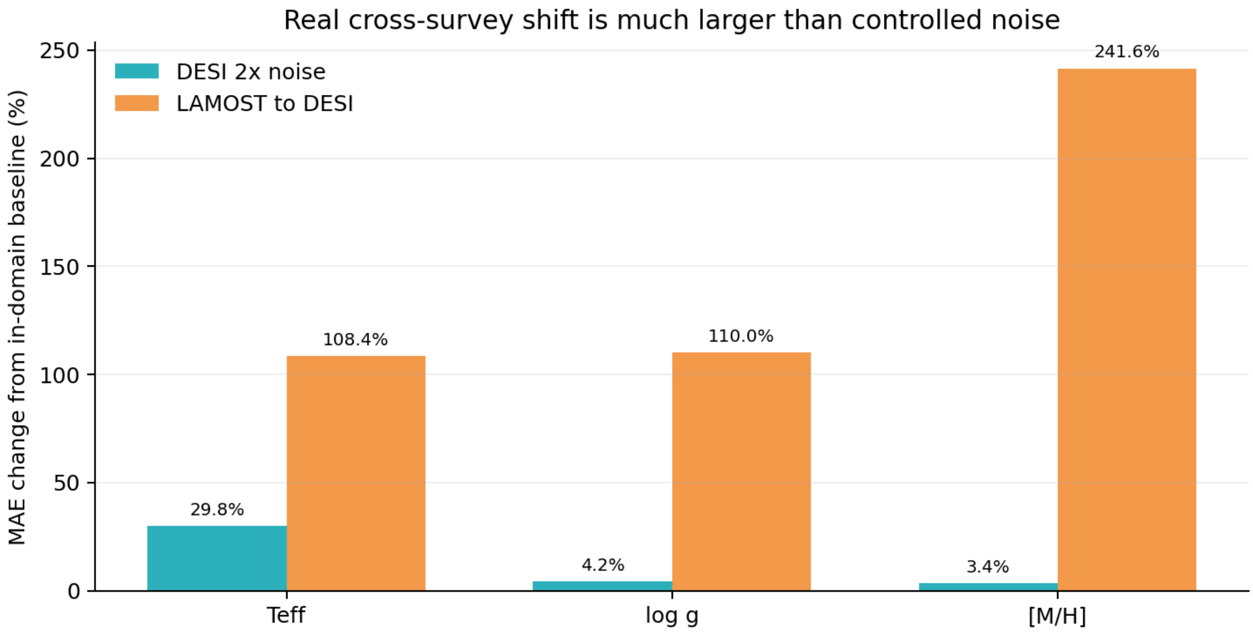
Version 1.2.3 independently resamples source-holdout and target-evaluation stars for the MAE-difference interval. The frozen public result directory predates source per-star prediction export, so its numerical two-sample interval is produced only when the non-redistributed survey contracts are rerun.

Shared contract

- Exact label scale: APOGEE DR12 ASPCAP v603 calibrated TEFF, LOGG, and PARAM_M_H.
- The legacy file field `feh` stores global [M/H], not elemental FE_H.
- Rest-frame 4000-5500 A on a common log-wavelength grid; Gaussian resolution matching to R=1800; no sharpening.
- Normalization, imputation, outlier thresholds, and 64-component PCA fit on source training only.
- Default estimator: separate 400-tree ExtraTrees regressors; target access is tagged for every method.

The gate was written before fitting and passed without changing a threshold after outcomes were observed. Input hashes and query metadata are frozen in the acquisition manifest.

2. Real survey shift versus controlled noise

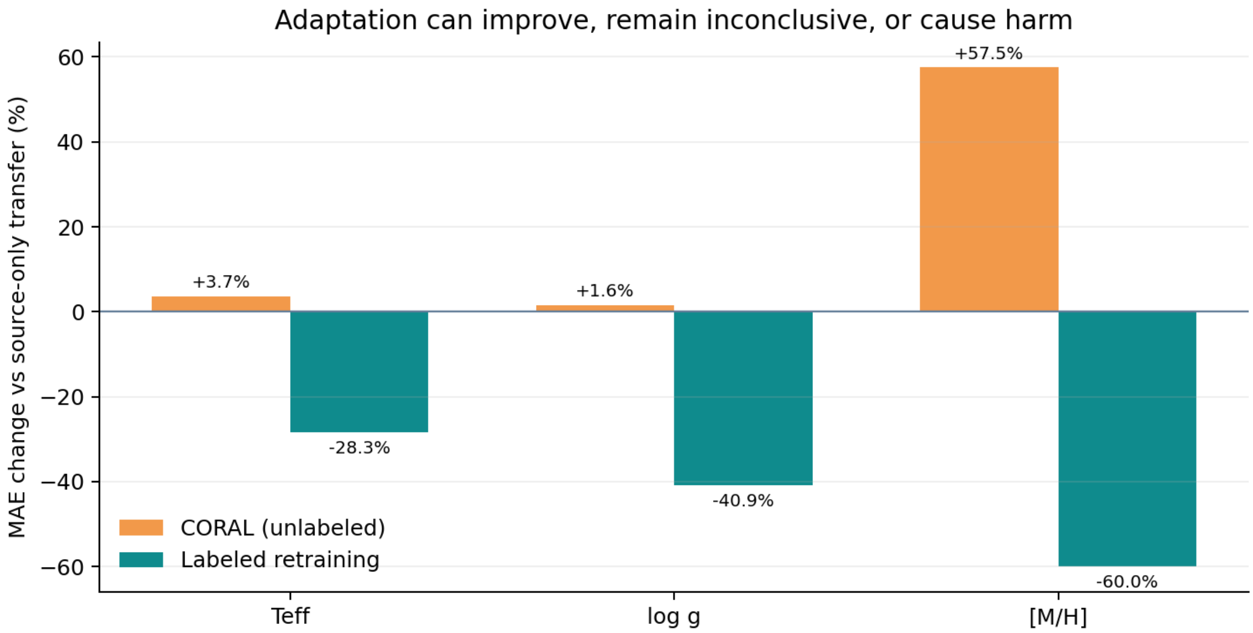


At 2x recorded noise in the controlled DESI experiment, temperature MAE rose by 29.8%, while gravity and metallicity changed by about 4%. The real LAMOST-to-DESI penalty is much larger for every target. This is a descriptive comparison, not a causal claim that the two shifts share a mechanism.

Target	LAMOST MAE	DESI MAE	Change	DESI MAE 95% CI	DESI R2
Teff	52.5 K	109.4 K	+108.4%	103.2-115.4 K	0.656
log g	0.124	0.260	+110.0%	0.240-0.279	0.558
[M/H]	0.073	0.250	+241.6%	0.228-0.270	0.220

The target errors are not just noisier: source-only predictions acquire positive biases of +44.6 K, +0.191 dex in log g, and +0.220 dex in [M/H]. Operational reliability therefore requires bias and calibration checks, not only a rank score.

3. Adaptation is method- and parameter-dependent

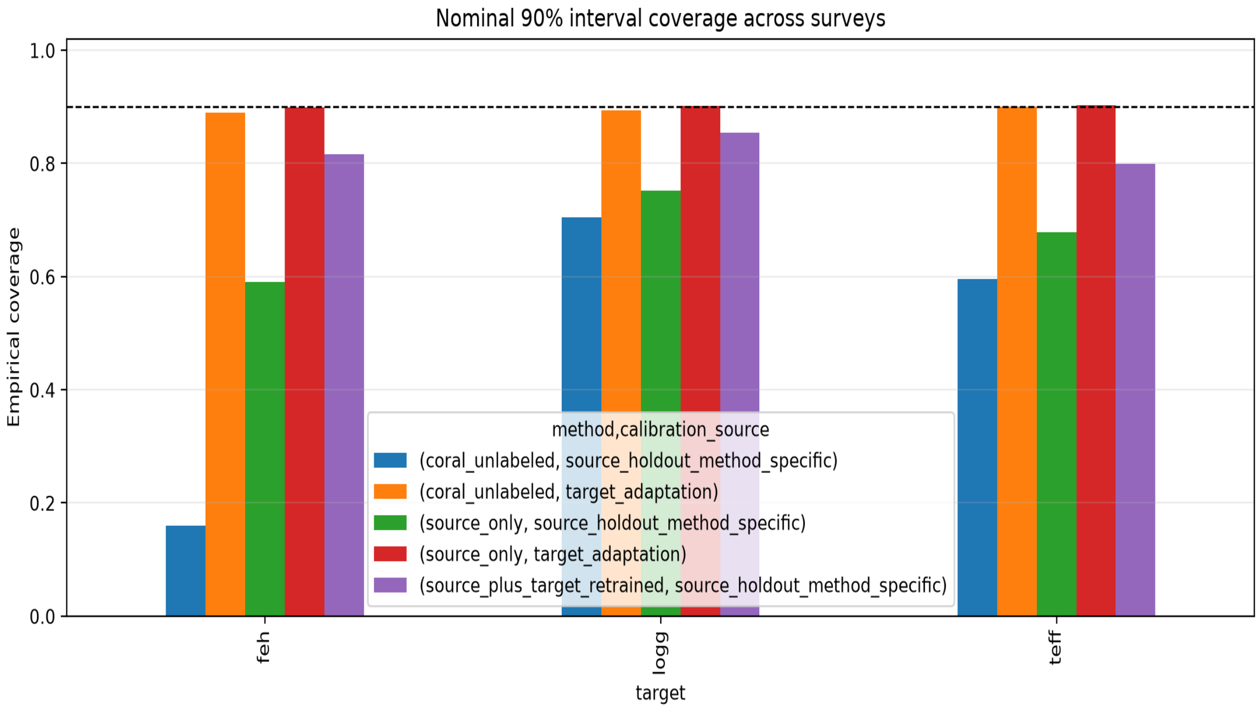


Method	Target access	Teff MAE	log g MAE	[M/H] MAE
Source only	none	109.4 K	0.260	0.250
CORAL	467 unlabeled spectra	113.5 K	0.264	0.393
Retraining	467 labeled stars	78.4 K	0.153	0.100

- CORAL Teff: +4.0 K paired difference, 95% CI -0.02 to +7.94 K - inconclusive, not equivalent.
- CORAL log g: +0.004 dex, 95% CI -0.004 to +0.013 - inconclusive.
- CORAL [M/H]: +0.144 dex, 95% CI +0.138 to +0.149 - detectable harm.
- Labeled retraining improves all three targets by 28.3%-60.0%, with paired intervals excluding zero.

This controlled three-way comparison supports a stronger conclusion than 'adaptation helps': target access, parameter, and method determine whether adaptation helps, does nothing detectable, or causes harm.

4. Uncertainty becomes overconfident under shift

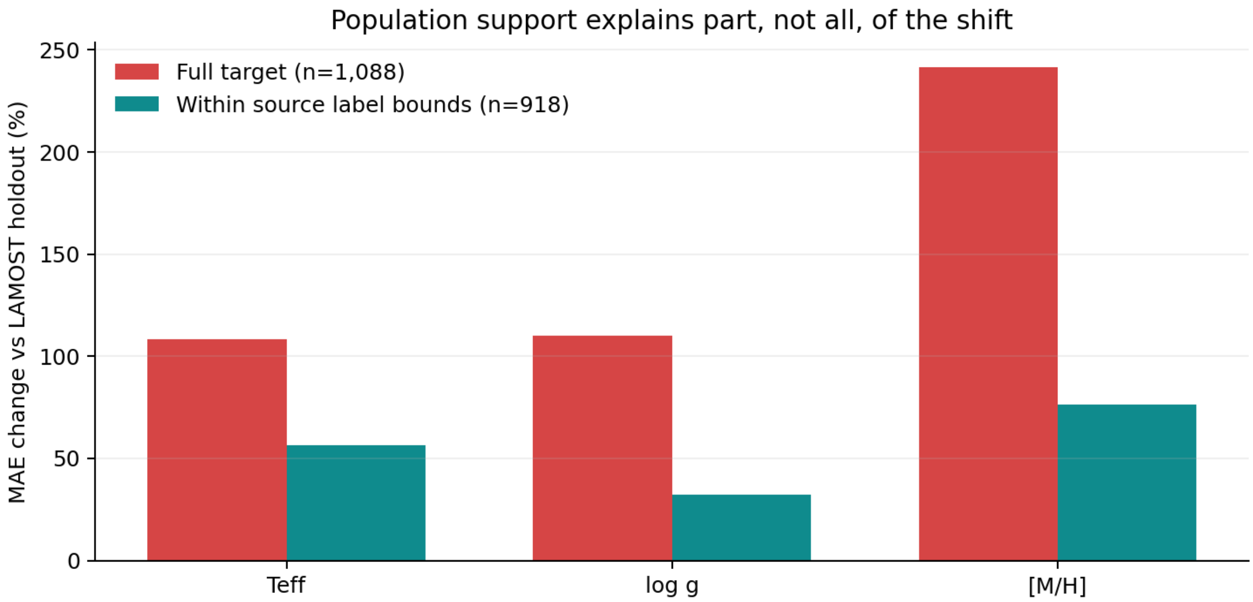


Nominal 90% intervals calibrated on LAMOST holdout residuals under-cover on DESI. The failure is especially severe for source-only [M/H] (59.0%) and CORAL [M/H] (16.0%). Calibration is therefore a distinct deployment failure even when point accuracy appears acceptable.

Method	Calibration data	Teff	log g	[M/H]
Source only	LAMOST holdout	67.8%	75.2%	59.0%
CORAL	LAMOST holdout	59.6%	70.5%	16.0%
Retraining	LAMOST holdout	80.0%	85.5%	81.6%
Source only	DESI adaptation	90.3%	90.2%	89.9%
CORAL	DESI adaptation	90.1%	89.3%	89.0%

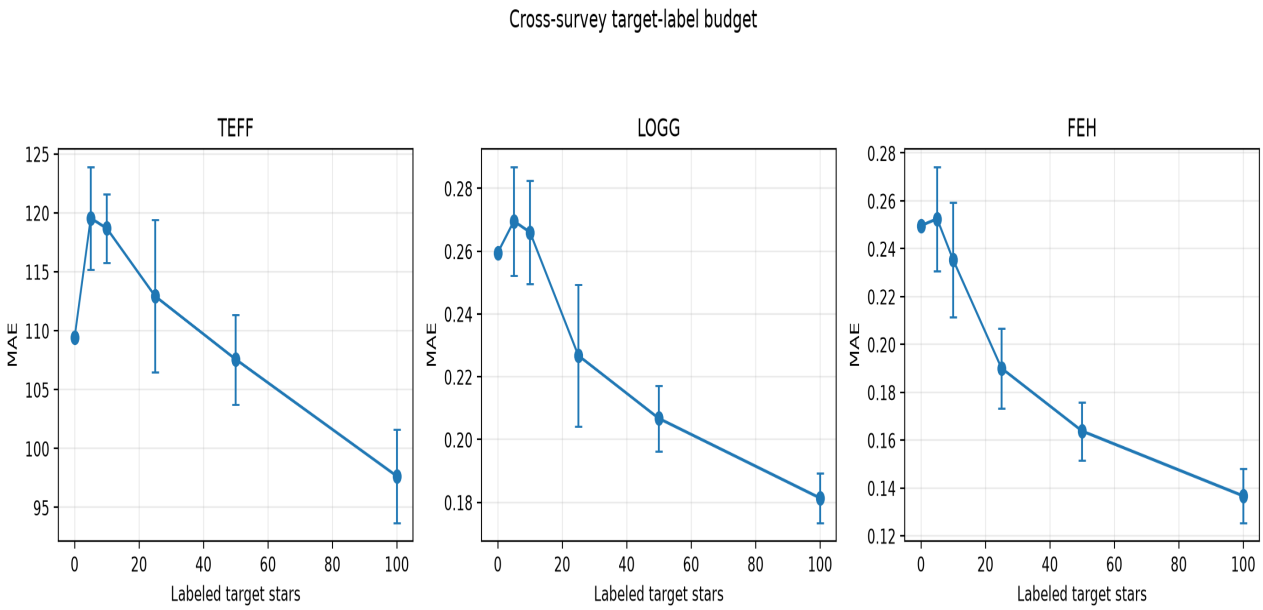
Target recalibration uses labels from the 467-star adaptation partition only; no target-evaluation label is touched. Coverage returns close to nominal, but interval widths expand substantially - honest uncertainty is not free.

5. Population support and label efficiency



The full target contains stellar labels outside the source-training range, including the metal-poor subgroup. Restricting to 918 stars jointly inside the source minima and maxima reduces degradation, but leaves a clear 32%-76% penalty. Population extrapolation matters; it does not explain the whole result.

Target-label budget



Across ten repeated draws, five target labels increase median error for every parameter. At 100 labels, median MAE improves by 11.6% for Teff, 31.1% for log g, and 46.0% for [M/H]. Label acquisition needs a budget curve, not a single convenient operating point.

6. Robustness, ablations, and physical checks

Formal reference-label errors

A paired 1,000-replicate sensitivity analysis perturbs APOGEE labels by their reported independent Gaussian one-sigma errors. Retraining remains better than source-only in every replicate for all targets; CORAL [M/H] remains worse in every replicate. Teff and log g CORAL differences remain inconclusive.

Model-family ablation

Family	Target Teff MAE	Target log g MAE	Target [M/H] MAE
Ridge	92.7 K	0.200	0.164
ExtraTrees	110.6 K	0.262	0.251
MLP	130.3 K	0.248	0.185

Ridge has the best absolute target performance in this ablation, while the small MLP beats ExtraTrees on log g and [M/H] but not Teff. No single family wins every parameter, and all show a substantial source-to-target penalty.

OOD and physical plausibility

- Keeping the 50% least-OOD source-only targets reduces MAE from 109.4 to 79.9 K, 0.260 to 0.163 dex, and 0.250 to 0.140 dex.
- No prediction violates the declared hard bounds for temperature, gravity, or metallicity.
- Isochrone-manifold consistency is explicitly not evaluated because no cited population-appropriate grid was supplied.

  Formal-error propagation is a sensitivity analysis, not independent validation of APOGEE. Hard bounds are necessary but weak physical tests; the unrun isochrone check remains a real limitation.

7. Reproducibility, limitations, and claim boundary

Reproducibility contract

- 51 deterministic tests; exact YAML configuration; immutable source and target SHA-256 hashes.
- Frozen prespecified quality gate and object-disjoint split manifest.
- Per-star predictions, star-bootstrap intervals, paired adaptation effects, calibration, label budgets, support, OOD, subgroups, ablations, and physical checks.
- Version-pinned acquisition URLs, Data Lab SQL, SPARCL response, duplicate rule, and public target selection.
- Raw survey spectra and processed spectral arrays excluded from the public archive; builders and hashes included.

Limitations

- The common APOGEE scale is consistent reference data, not independent absolute truth; shared ASPCAP systematics may remain.
- The primary effect combines instrument, reduction, S/N, and population/covariate shift. The 918-star support analysis is narrower, not perfectly causal.
- The Ho source is high-S/N and has no stars below [M/H] = -1.5 after quality cuts; the DESI target is giant- and backup-program-dominated.
- Resolution matching uses a Gaussian approximation, not exact per-spectrum DESI/LAMOST line-spread matrices.
- No B+R+Z fusion, external-label replication, or isochrone-manifold result is claimed.

Claim boundary

Supported: a large, calibrated, object-disjoint LAMOST DR2 to DESI DR1 transfer penalty on the APOGEE DR12 scale; method-specific adaptation and calibration findings; label-budget, support, OOD, subgroup, ablation, formal-error-sensitivity, and hard-bound results for the frozen data.

Not supported: pure instrument causality, independent validation of APOGEE, universal cross-survey generalization, exact LSF conclusions, isochrone consistency, or a DOI before deposition.

References and public data

Ho et al. (2017), ApJ 836, 5, doi:10.3847/1538-4357/836/1/5. The Cannon LAMOST tutorial: annayqho.github.io/TheCannon/lamost_tutorial.html. APOGEE DR12 allStar-v603: sdss4.org/dr12/data_access/bulk-data-downloads/. DESI DR1: data.desi.lbl.gov/doc/releases/dr1/. NOIRLab DESI/SPARCL: datalab.noirlab.edu/data/desi.